# #Hands-On Activity 10 | Explaining Visually to Influence with Data

Name : Benj Federic C. De Leon
Course Code and Title : CPE 031- Visualizations and Data Analysis
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Instructor : Engr. Jimlord M. Quejado

## 1. Objectives:

This activity aims to help students demonstrate how to influence and communicate effectively with data by identifying audience needs, crafting a clear message, and telling a story visually through charts.

## 2. Intended Learning Outcomes (ILOs):

By the end of this activity, students should be able to:

1. Analyze the audience to understand their level of data knowledge and needs.
2. Evaluate how well a message and visualization communicate insights.
3. Create a clear and meaningful data story using Python visualization tools.

## 3. Discussions:

Data storytelling goes beyond showing charts it’s about influencing decisions and sparking understanding.

Know Your Audience: Decision-makers may prefer clear summaries, while analysts like detailed data.

Know Your Message: Focus on the why it matters part of the story.

Tell a Story: Use preattentive attributes color, size, and position to guide attention.

Structure Your Story: Like Aristotle’s rhetorical triangle, combine logic (logos), emotion (pathos), and credibility (ethos).

## 4. Procedures:

Step 1. Import Libraries and Create Sample Data

In [38]:
```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd

# Sample sales dataset
data = {
    "Month": ["Jan", "Feb", "Mar", "Apr", "May", "Jun"],
    "Sales": [200, 250, 180, 300, 280, 350],
    "Marketing_Spend": [50, 60, 45, 70, 65, 80]
}
df = pd.DataFrame(data)
df
```

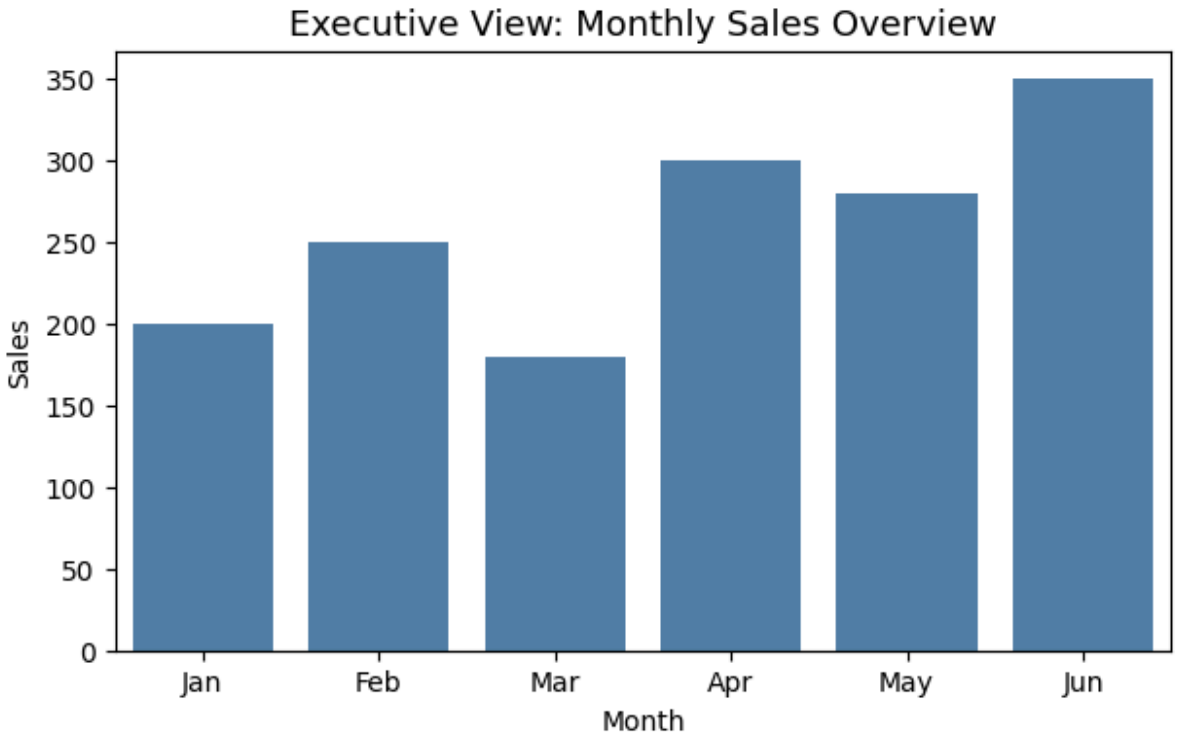
Out[38]:

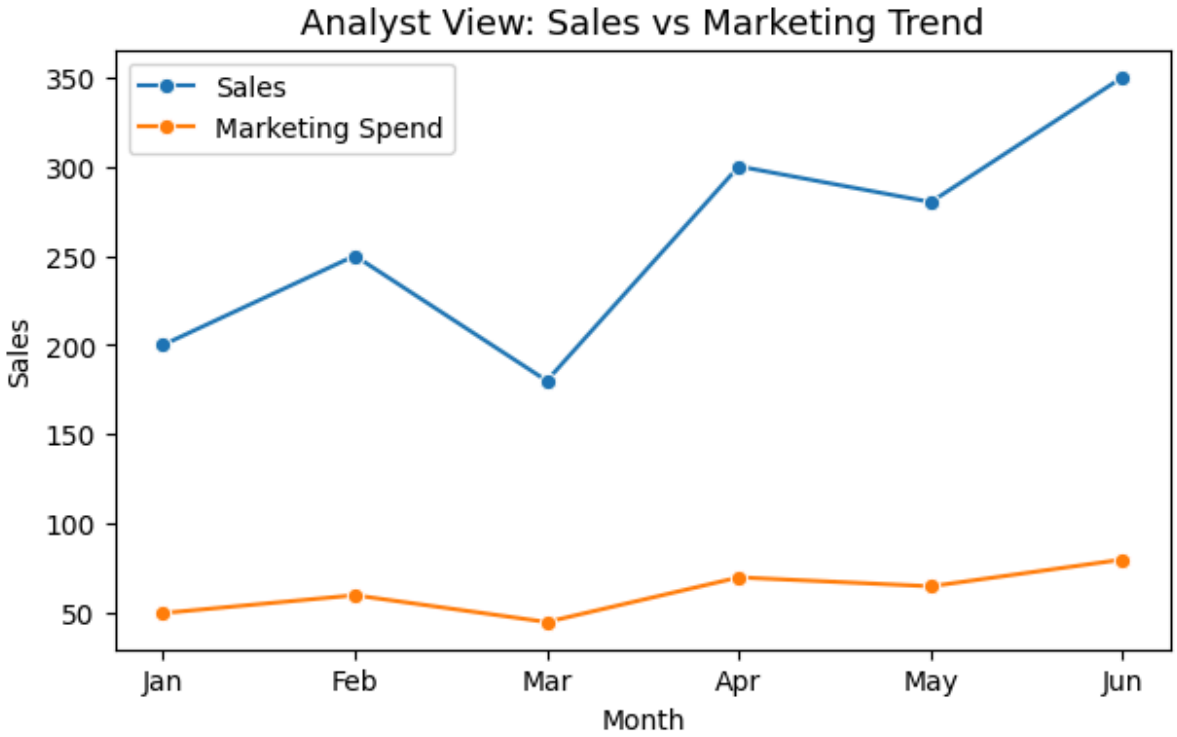
| | Month | Sales | Marketing_Spend |
|---|---|---|---|
| **0** | Jan | 200 | 50 |
| **1** | Feb | 250 | 60 |
| **2** | Mar | 180 | 45 |
| **3** | Apr | 300 | 70 |
| **4** | May | 280 | 65 |
| **5** | Jun | 350 | 80 |

## Step 2. Know Your Audience — Compare Chart Styles

In [39]:
```
# Executive audience: prefer clarity (simple bar)
plt.figure(figsize=(7,4))
sns.barplot(x="Month", y="Sales", data=df, color="steelblue")
plt.title("Executive View: Monthly Sales Overview", fontsize=13)
plt.show()

# Analyst audience: prefer detailed trends (line chart)
plt.figure(figsize=(7,4))
sns.lineplot(x="Month", y="Sales", data=df, marker="o", label="Sales")
sns.lineplot(x="Month", y="Marketing_Spend", data=df, marker="o",
label="Marketing Spend")
plt.title("Analyst View: Sales vs Marketing Trend", fontsize=13)
plt.legend()
plt.show()
```





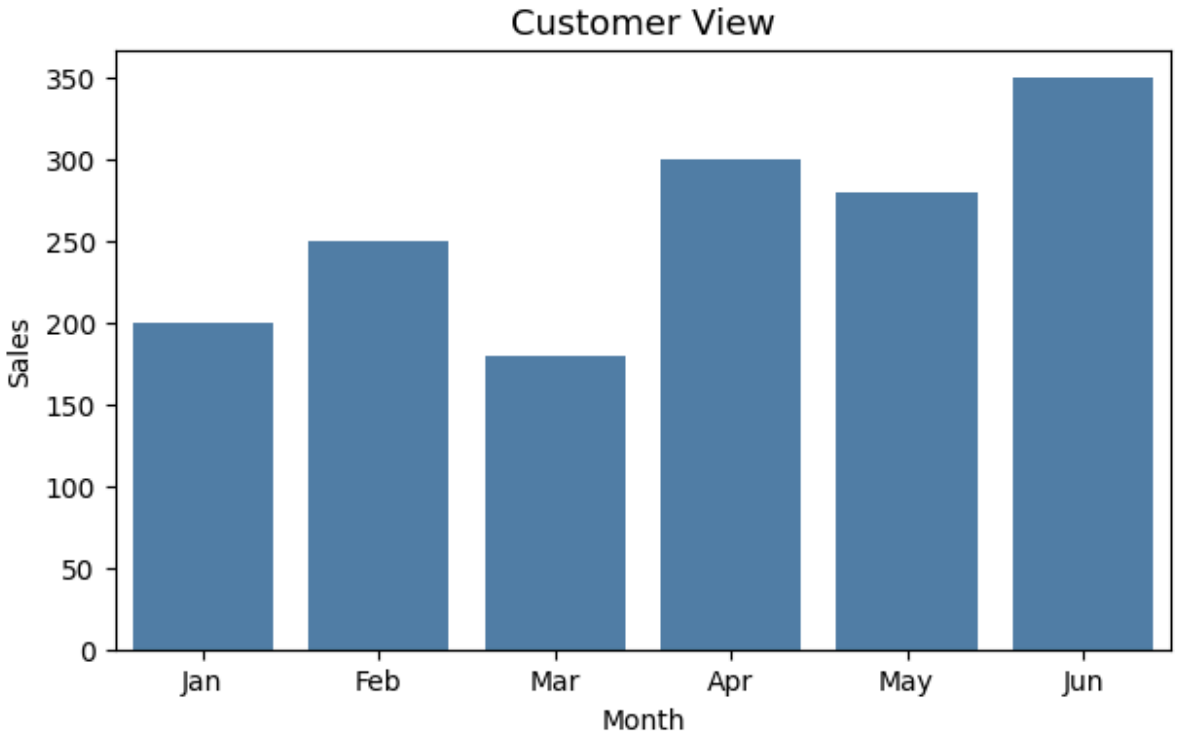

**Task 1:** Create one additional visualization suited for a different audience (e.g., a customer, investor, or teammate).

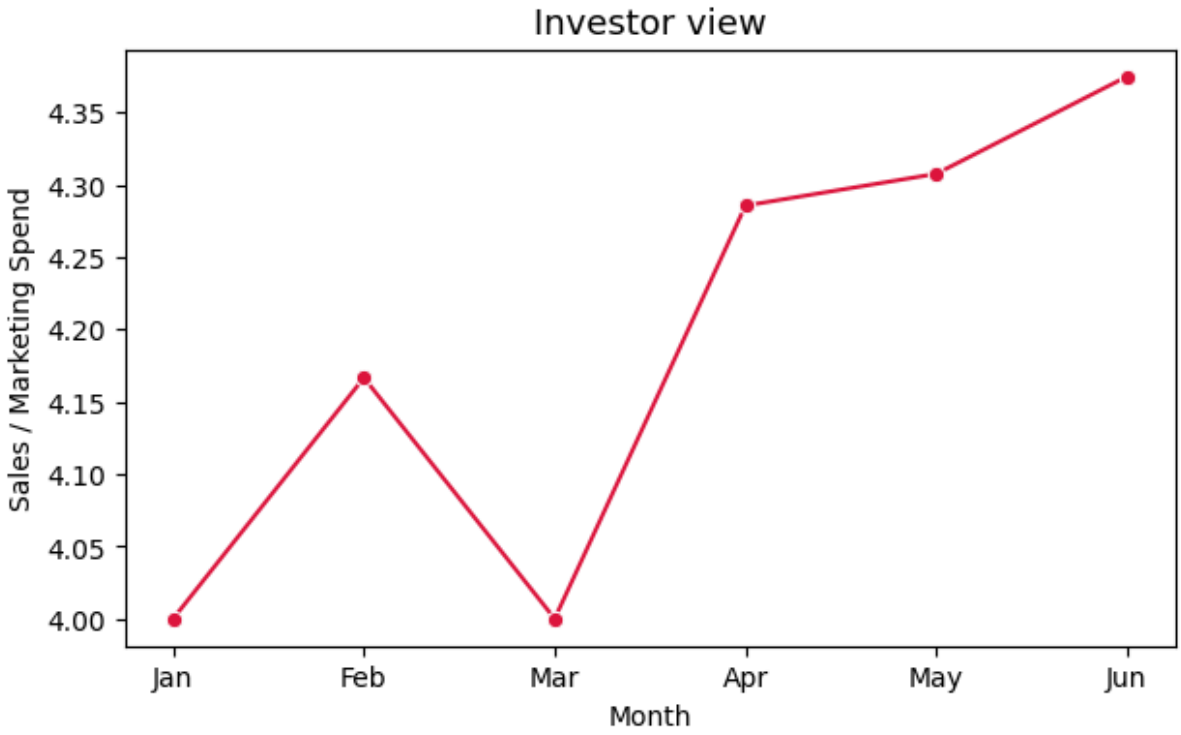
In [40]:

```
plt.figure(figsize=(7,4))
sns.barplot(x="Month", y="Sales", data=df, color="steelblue")
plt.title("Customer View", fontsize=13)
plt.ylabel("Sales")
plt.show()

# Investor audience: return on marketing
df["Sales_per_Marketing"] = df["Sales"] / df["Marketing_Spend"]

plt.figure(figsize=(7,4))
sns.lineplot(x="Month", y="Sales_per_Marketing", data=df, marker="o", color=
"crimson")
plt.title("Investor view", fontsize=13)
plt.ylabel("Sales / Marketing Spend")
plt.show()
```

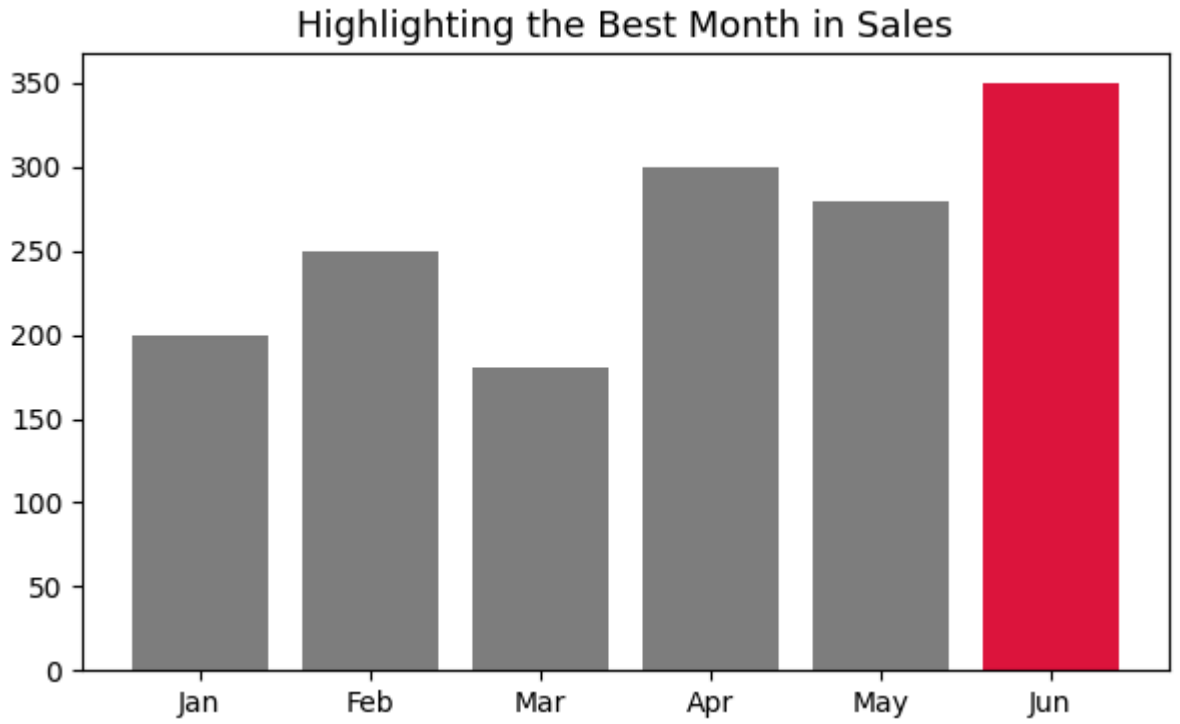





Step 3. Know Your Message — Highlight What Matters

In [41]:
```
# Highlight the highest sales month (preattentive color cue)
highlight = df["Sales"].idxmax()

colors = ["gray" if i != highlight else "crimson" for i in range(len(df))]
plt.figure(figsize=(7,4))
plt.bar(df["Month"], df["Sales"], color=colors)
plt.title("Highlighting the Best Month in Sales", fontsize=13)
plt.show()
```



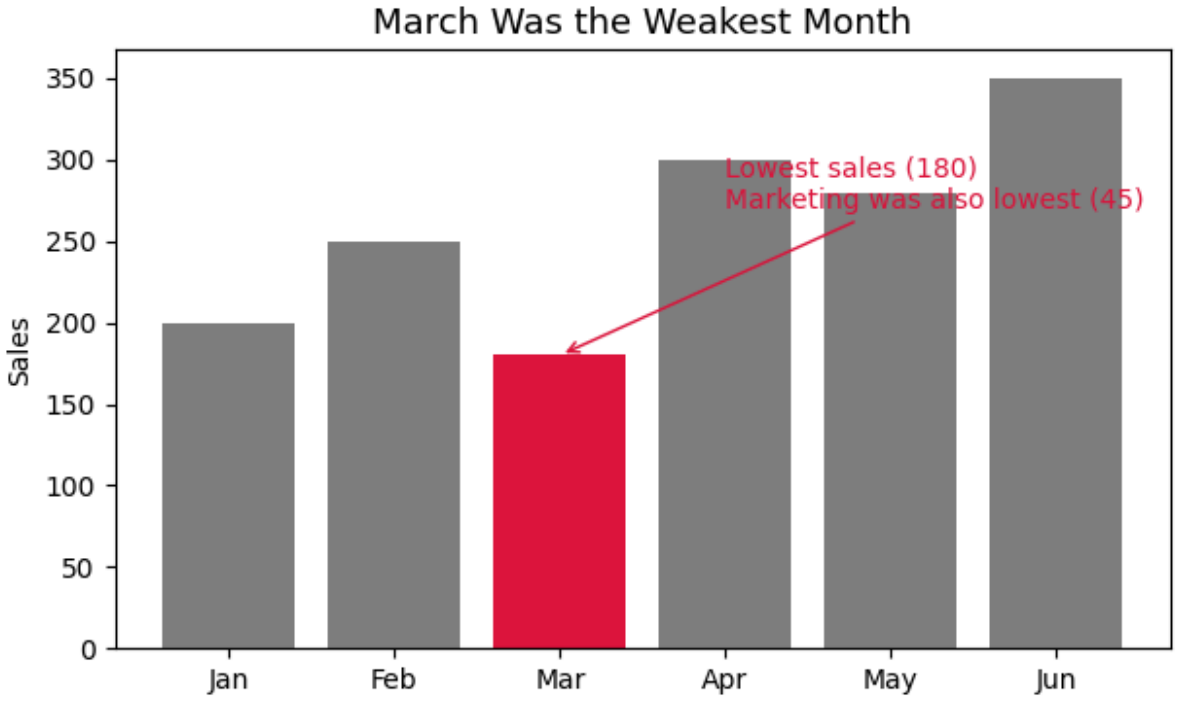
**Task 2:** Use color or annotations to emphasize a key insight (e.g., lowest point, sudden increase).

In [47]:
```
# Highlight the highest sales month
lowest = df["Sales"].idxmin()

colors = ["gray" if i != lowest else "crimson" for i in range(len(df))]
plt.figure(figsize=(7,4))
plt.bar(df["Month"], df["Sales"], color=colors)

plt.annotate("Lowest sales (180)\nMarketing was also lowest (45)",
             xy=(lowest, df["Sales"][lowest]),
             xytext=(lowest + 1, df["Sales"][lowest] + 90),
             arrowprops=dict(arrowstyle="→", color="crimson"),
             fontsize=10, color="crimson")

plt.title("March Was the Weakest Month", fontsize=13)
plt.ylabel("Sales")
plt.show()
```

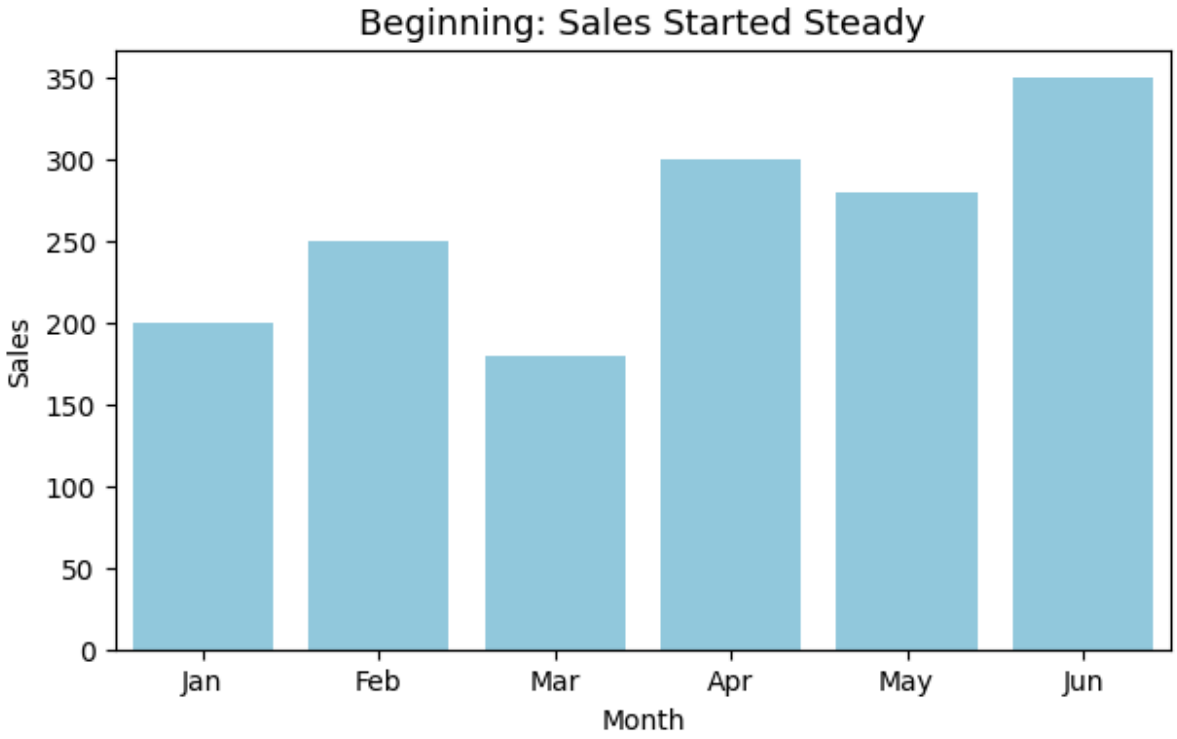


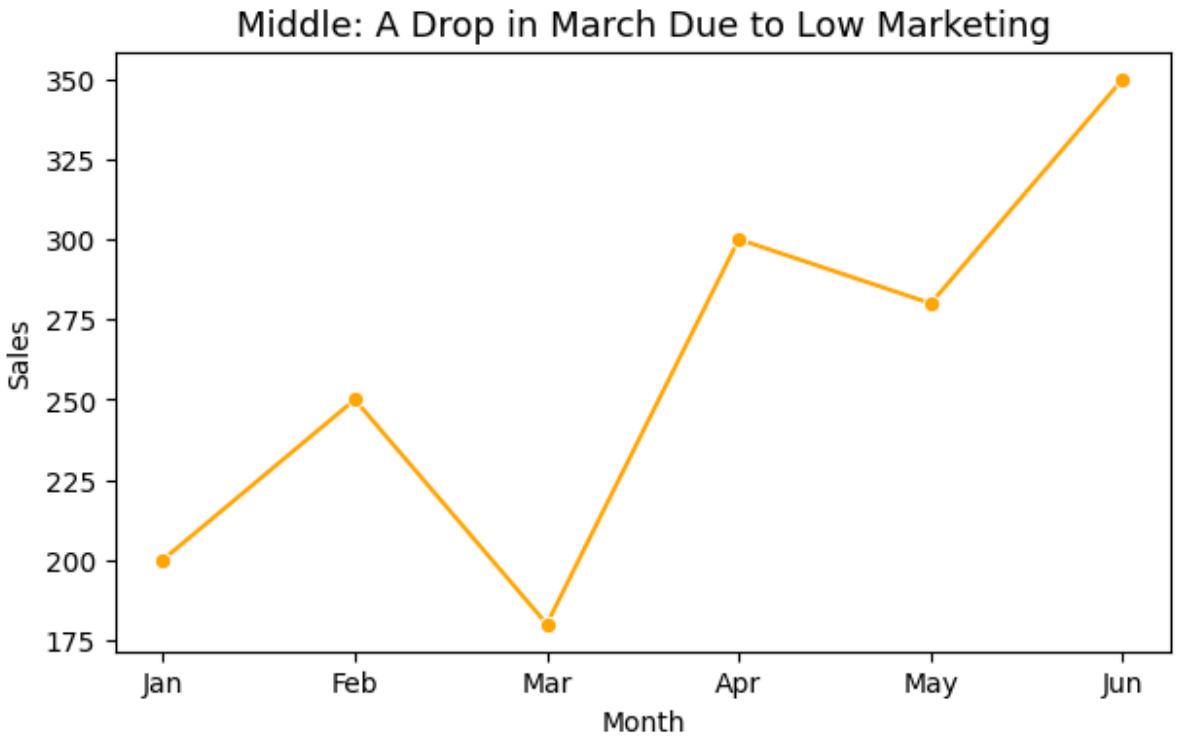

Step 4. Storytelling with Charts — Beginning, Middle, End

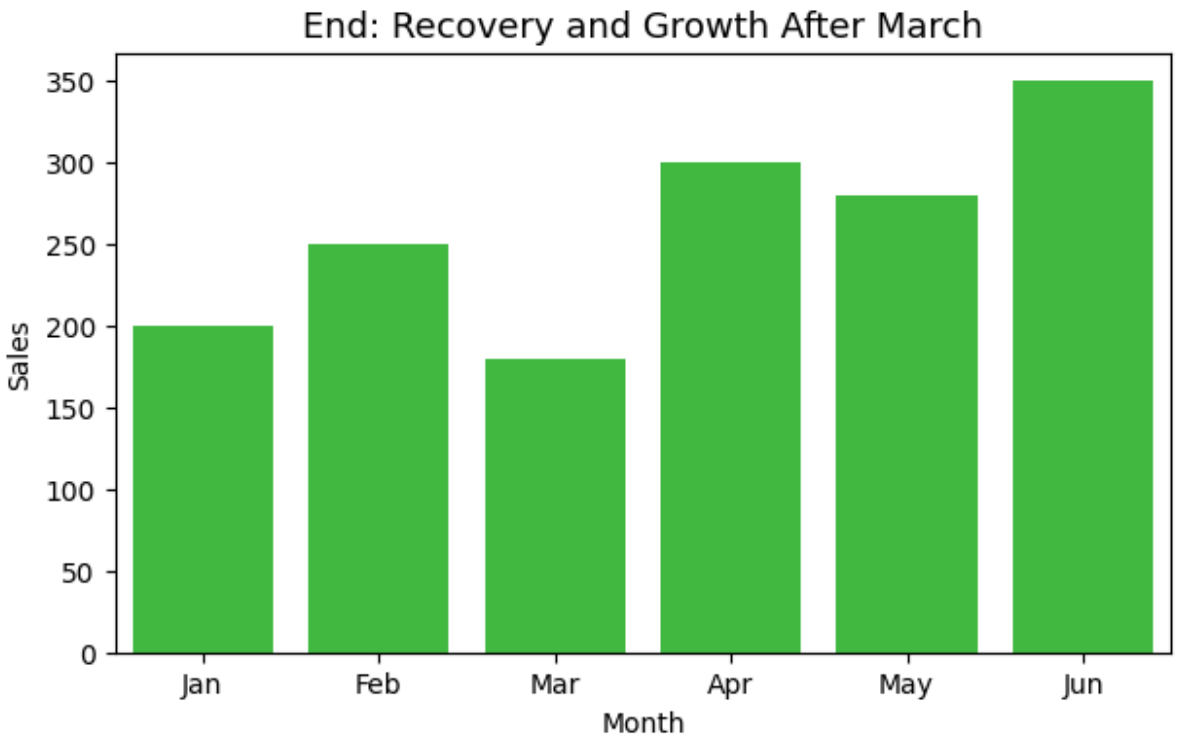
In [43]:
```
# Beginning: Context
plt.figure(figsize=(7,4))
sns.barplot(x="Month", y="Sales", data=df, color="skyblue")
plt.title("Beginning: Sales Started Steady", fontsize=13)
plt.show()

# Middle: Change or Challenge
plt.figure(figsize=(7,4))
sns.lineplot(x="Month", y="Sales", data=df, color="orange", marker="o")
plt.title("Middle: A Drop in March Due to Low Marketing", fontsize=13)
plt.show()

# End: Resolution
plt.figure(figsize=(7,4))
sns.barplot(x="Month", y="Sales", data=df, color="limegreen")
plt.title("End: Recovery and Growth After March", fontsize=13)
plt.show()
```





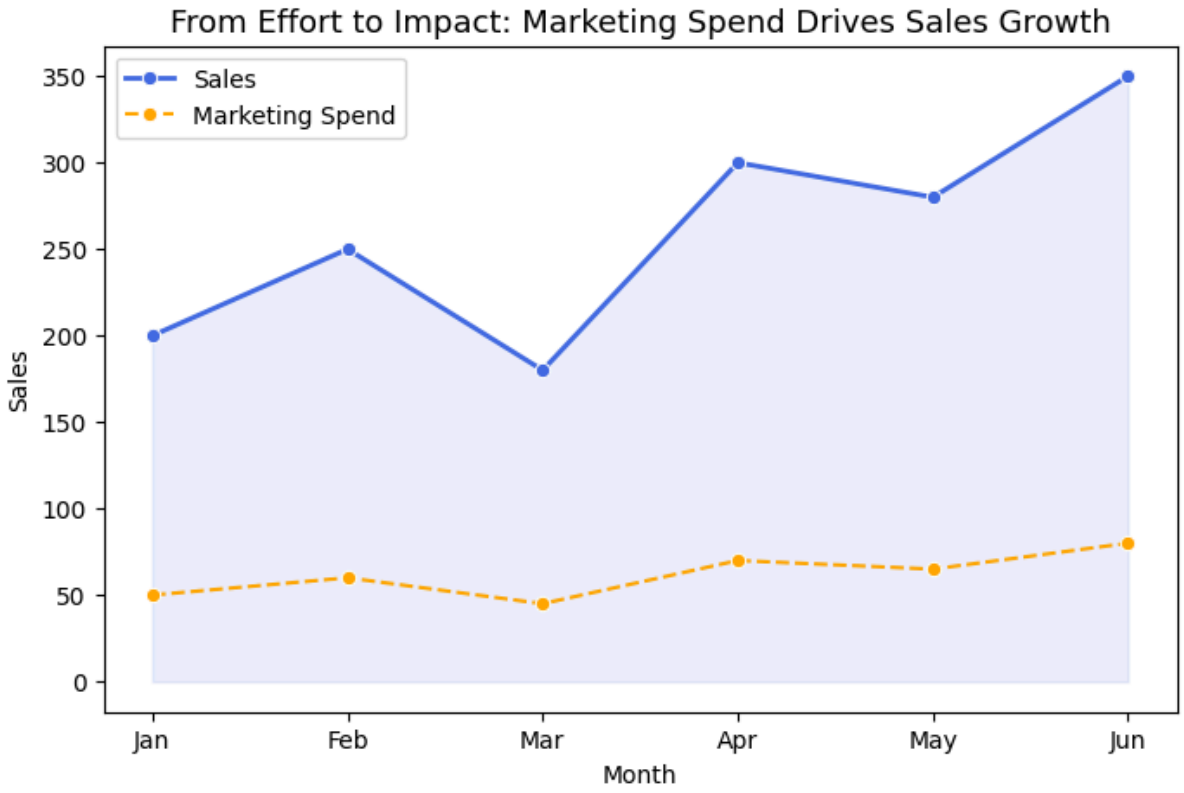

**Task 3:** Write one or two lines describing your data story from beginning → middle → end.

**Answer:** Sales started steady in January and February (Beginning). In March, sales dropped to 180 when marketing spend was also at its lowest (Middle). After marketing spend increased again, sales recovered and reached a peak of 350 in June (End), showing that higher marketing effort goes together with higher sales.

Step 5. Bringing It All Together — Presentation Design

In [44]:
```
# Simple combined story chart
plt.figure(figsize=(8,5))
sns.lineplot(x="Month", y="Sales", data=df, marker="o", linewidth=2,
color="royalblue", label="Sales")
sns.lineplot(x="Month", y="Marketing_Spend", data=df, marker="o",
linestyle="--", color="orange", label="Marketing Spend")

plt.fill_between(df["Month"], df["Sales"], alpha=0.1, color="royalblue")
plt.title("From Effort to Impact: Marketing Spend Drives Sales Growth",
fontsize=13)
plt.legend()
plt.show()
```




**5. Supplementary Activity:**

1. Use a small dataset available online (kaggle, google dataset, etc.).
2. Visualize it using a bar chart.
3. Highlight one important data point (e.g., the highest or lowest value) using a distinct color.
4. Add a title and annotation that tell the story behind the data.

**Supplementary Activity: Sleep Health and Lifestyle Dataset (Kaggle)**
***Dataset:*** *https://www.kaggle.com/datasets/uom190346a/sleep-health-and-lifestyle-dataset*

In [45]:
```
sleep_df =
pd.read_csv("/content/HoA10p1_Sleep_health_and_lifestyle_dataset.csv")
sleep_df.head()
```

Out[45]: Columns 1–8 of 14

| | Person ID | Gender | Age | Occupation | Sleep Duration | Quality of Sleep | Physical Activity Level |
|---|---|---|---|---|---|---|---|
| **0** | 1 | Male | 27 | Software Engineer | 6.1 | 6 | 42 |
| **1** | 2 | Male | 28 | Doctor | 6.2 | 6 | 60 |
| **2** | 3 | Male | 28 | Doctor | 6.2 | 6 | 60 |
| **3** | 4 | Male | 28 | Sales Representative | 5.9 | 4 | 30 |
| **4** | 5 | Male | 28 | Sales Representative | 5.9 | 4 | 30 |

Columns 9–14 of 14 · first column repeated

| | Stress Level | BMI Category | Blood Pressure | Heart Rate | Daily Steps | Sleep Disorder |
|---|---|---|---|---|---|---|
| **0** | 6 | Overweight | 126/83 | 77 | 4200 | NaN |
| **1** | 8 | Normal | 125/80 | 75 | 10000 | NaN |
| **2** | 8 | Normal | 125/80 | 75 | 10000 | NaN |
| **3** | 8 | Obese | 140/90 | 85 | 3000 | Sleep Apnea |
| **4** | 8 | Obese | 140/90 | 85 | 3000 | Sleep Apnea |

In [46]:

```
#Average sleep quality for each occupation
avg_sleep = sleep_df.groupby("Occupation")["Quality of
Sleep"].mean().sort_values().reset_index()
avg_sleep

#Highlight the lowest value using a distinct color
lowest = avg_sleep["Quality of Sleep"].idxmin()
colors = ["lightgray" if i != lowest else "crimson" for i in
range(len(avg_sleep))]

plt.figure(figsize=(9,5))
plt.bar(avg_sleep["Occupation"], avg_sleep["Quality of Sleep"],
color=colors)

#Annotation that tells the story behind the data
worst_job = avg_sleep["Occupation"][lowest]
worst_score = avg_sleep["Quality of Sleep"][lowest]
plt.annotate(f"{worst_job} has the\nlowest sleep quality
({worst_score:.1f}/10)",
             xy=(lowest, worst_score),
             xytext=(lowest + 1.5, worst_score + 0.6),
             arrowprops=dict(arrowstyle="→", color="crimson"),
             fontsize=10, color="crimson")
plt.title("Which Occupation Sleeps the Worst? Average Sleep Quality by
Occupation", fontsize=13)
plt.xlabel("Occupation")
plt.ylabel("Average Quality of Sleep (1-10)")
plt.xticks(rotation=45, ha="right")
plt.ylim(0, 10)
plt.tight_layout()
plt.show()
```




## 6. Conclusion/Learnings/Analysis:

- **Learnings**: different audiences need different visuals: executives want simple bar charts, while analysts prefer detailed line charts. Adding colors and labels helps people spot key takeaways immediately, like the highest or lowest points, without having to search. Framing data with a clear beginning, middle, and end turns plain numbers into a convincing story that guides better decisions.
- **Analysis**: Sales Representatives have the worst sleep quality at 4.0 out of 10, while Engineers get the best sleep at 8.4 out of 10. Overall, jobs with steady office routines like engineers, lawyers, and accountants report much better sleep than high-stress or sales roles.